

D2: Programming Project

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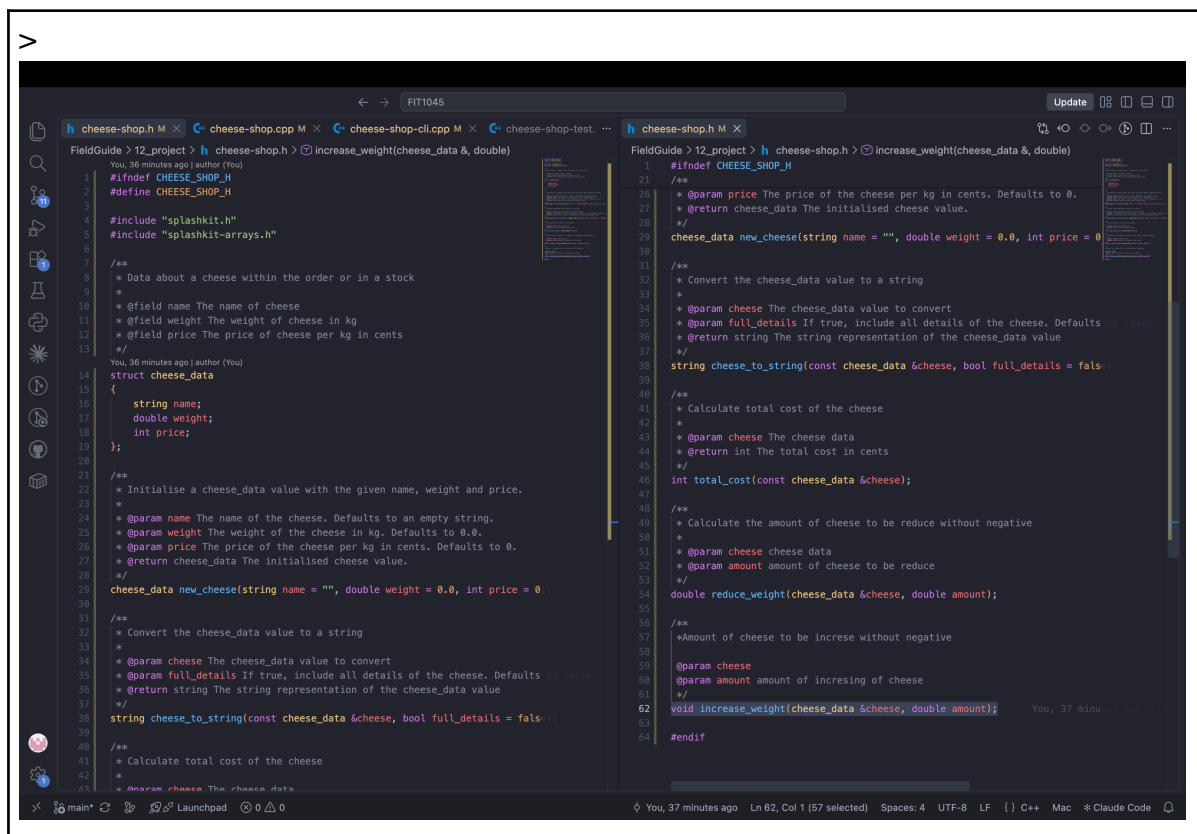
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Overview

This task is an opportunity for you to demonstrate your ability to work on a larger programming project using an iterative process.

Do the following:

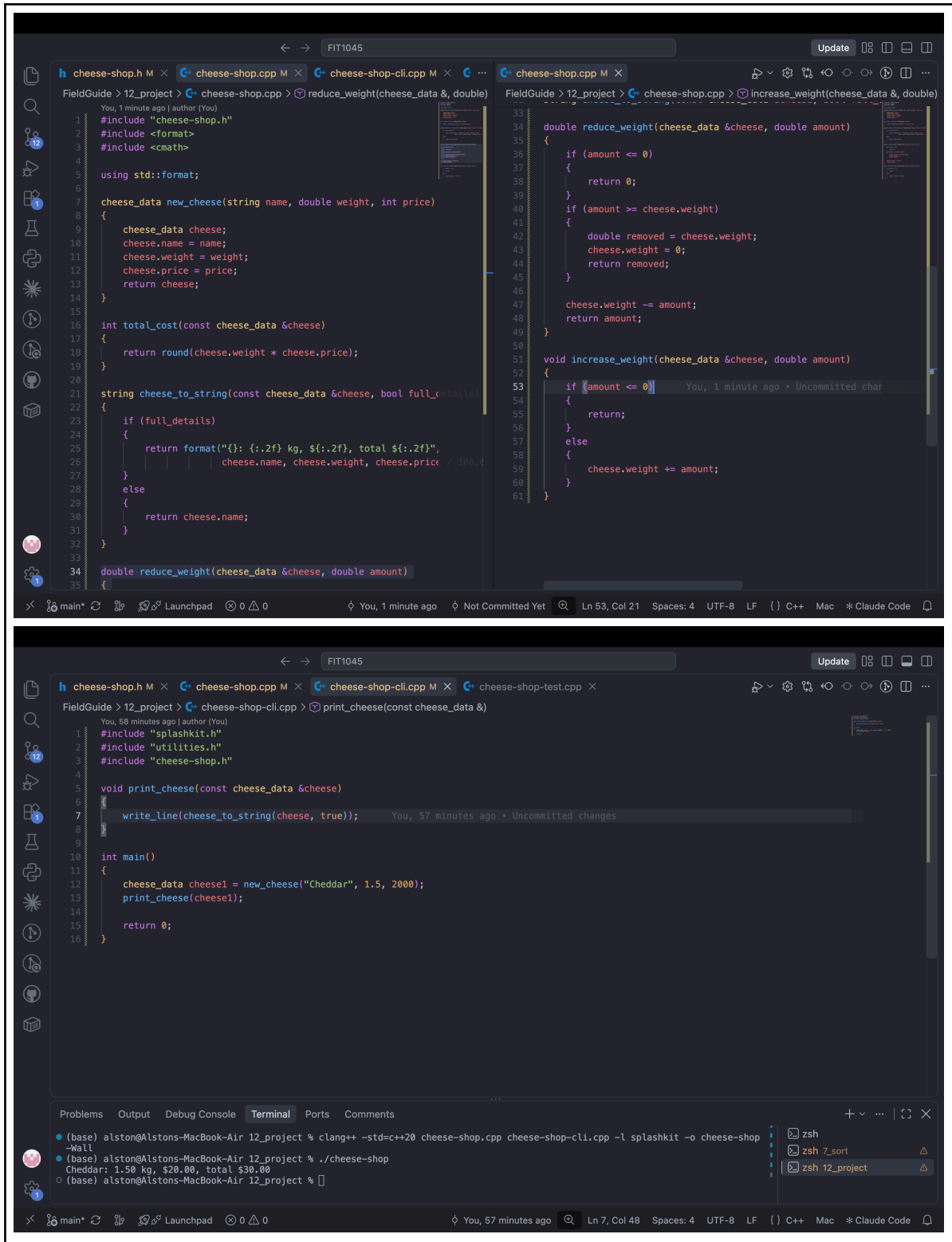
- Everything you need is in the [Project](#) chapter of our [Programmer's Field Guide](#). Make use of the [guided tours](#) to explore these concepts.
- Add a screenshot showing your progress through [iteration 1](#). Show the program running and some sample output.



```

1  #include <iostream>
2  #include <string>
3  #include <string.h>
4  #include "splashkit.h"
5  #include "splashkit-arrays.h"
6
7  /**
8   * Data about a cheese within the order or in a stock
9   *
10  * @field name The name of cheese
11  * @field weight The weight of cheese in kg
12  * @field price The price of cheese per kg in cents
13  */
14  struct cheese_data
15  {
16      string name;
17      double weight;
18      int price;
19  };
20
21  /**
22  * Initialise a cheese_data value with the given name, weight and price.
23  *
24  * @param name The name of the cheese. Defaults to an empty string.
25  * @param weight The weight of the cheese in kg. Defaults to 0.0.
26  * @param price The price of the cheese per kg in cents. Defaults to 0.
27  * @return cheese_data The initialised cheese value.
28  */
29  cheese_data new_cheese(string name = "", double weight = 0.0, int price = 0)
30  {
31      /**
32       * Convert the cheese_data value to a string
33       *
34       * @param cheese The cheese_data value to convert
35       * @param full_details If true, include all details of the cheese. Defaults to false.
36       * @return string The string representation of the cheese_data value
37       */
38      string cheese_to_string(const cheese_data &cheese, bool full_details = false)
39      {
40          /**
41           * Calculate total cost of the cheese
42           *
43           * @param cheese The cheese data
44           * @return int The total cost in cents
45           */
46          int total_cost(const cheese_data &cheese);
47
48          /**
49           * Calculate the amount of cheese to be reduce without negative
50           *
51           * @param cheese cheese data
52           * @param amount amount of cheese to be reduce
53           */
54          double reduce_weight(cheese_data &cheese, double amount);
55
56          /**
57           * Amount of cheese to be incrise without negative
58           *
59           * @param cheese
60           * @param amount amount of incrising of cheese
61           */
62          void increase_weight(cheese_data &cheese, double amount);
63
64          #endif

```



The first screenshot shows the implementation of the `cheese_data` struct and functions `new_cheese`, `total_cost`, `cheese_to_string`, and `reduce_weight` in `cheese-shop.cpp`. The second screenshot shows the `print_cheese` function and the `main` function in `cheese-shop-cli.cpp`, which demonstrates the program's output.

```

1 #include "cheese-shop.h"
2 #include <format>
3 #include <cmath>
4
5 using std::format;
6
7 cheese_data new_cheese(string name, double weight, int price)
8 {
9     cheese_data cheese;
10    cheese.name = name;
11    cheese.weight = weight;
12    cheese.price = price;
13    return cheese;
14 }
15
16 int total_cost(const cheese_data &cheese)
17 {
18     return round(cheese.weight * cheese.price);
19 }
20
21 string cheese_to_string(const cheese_data &cheese, bool full_details)
22 {
23     if (full_details)
24     {
25         return format("{}: {:.2f} kg, ${:.2f}, total ${:.2f}",
26                        cheese.name, cheese.weight, cheese.price, 100.0);
27     }
28     else
29     {
30         return cheese.name;
31     }
32 }
33
34 double reduce_weight(cheese_data &cheese, double amount)
35 {
36     if (amount <= 0)
37     {
38         return 0;
39     }
40     if (amount >= cheese.weight)
41     {
42         double removed = cheese.weight;
43         cheese.weight = 0;
44         return removed;
45     }
46     cheese.weight -= amount;
47     return amount;
48 }
49
50 void increase_weight(cheese_data &cheese, double amount)
51 {
52     if (amount <= 0)
53     {
54         return;
55     }
56     else
57     {
58         cheese.weight += amount;
59     }
60 }
61

```

```

1 #include "splashkit.h"
2 #include "utilities.h"
3 #include "cheese-shop.h"
4
5 void print_cheese(const cheese_data &cheese)
6 {
7     write_line(cheese_to_string(cheese, true));
8 }
9
10 int main()
11 {
12     cheese_data cheese1 = new_cheese("Cheddar", 1.5, 2000);
13     print_cheese(cheese1);
14     return 0;
15 }
16

```

```

(base) alston@Alstons-MacBook-Air 12_project % clang++ -std=c++20 cheese-shop.cpp cheese-shop-cli.cpp -l splashkit -o cheese-shop
-Wall
(base) alston@Alstons-MacBook-Air 12_project % ./cheese-shop
Cheddar: 1.50 kg, $20.00, total $30.00
(base) alston@Alstons-MacBook-Air 12_project %

```

- Add a screenshot showing your progress through [iteration 2](#). Show the program running and some sample output.

FieldGuide > 12_project > h cheese-shop.h

```

1 #ifndef CHEESE_SHOP_H
2 #define CHEESE_SHOP_H
3
4 #include "sglaskit.h"
5 #include "sglaskit-arrays.h"
6
7 /**
8  * Data about a cheese within the order or in a stock
9  *
10  * @field name The name of cheese
11  * @field weight The weight of cheese in kg
12  * @field price The price of cheese per kg in cents
13  */
14 You, 6 hours ago (author You)
15 struct cheese_data
16 {
17     string name;
18     double weight;
19     int price;
20 };
21
22 /**
23  * Data about the cheese shop
24  *
25  * @field cheese The list of cheese in shop
26  */
27 You, 6 hours ago (author You)
28 struct shop_data
29 {
30     dynamic_array<cheese_data> cheeses;
31 };
32
33 /**
34  * Initialise a cheese_data value with the given name, weight and price.
35  *
36  * @param name The name of the cheese. Defaults to an empty string.
37  * @param weight The weight of the cheese in kg. Defaults to 0.0.
38  * @param price The price of the cheese per kg in cents. Defaults to 0.
39  * @return string The initialised cheese value.
40  */
41 cheese_data
42 new_cheese(string name = "", double weight = 0.0, int price = 0);
43
44 /**
45  * Convert the cheese_data value to a string
46  *
47  * @param cheese The cheese_data value to convert
48  * @param full_details If true, include all details of the cheese. Defaults to false
49  * @return string The string representation of the cheese_data value
50  */
51 string cheese_to_string(const cheese_data &cheese, bool full_details = false);
52
53 /**
54  * Calculate total cost of the cheese
55  *
56  * @param cheese The cheese_data
57  * @return int The total cost in cents
58  */
59 int total_cost(const cheese_data &cheese);
60
61 /**
62  * Calculate the amount of cheese to be reduce without negative
63  *
64  * @param cheese The cheese_data
65  * @param amount amount of cheese to be reduce
66  */
67 double reduce_weight(cheese_data &cheese, double amount);
68
69 /**
70  * Amount of cheese to be increase without negative
71  *
72  * @param cheese The cheese_data
73  * @param amount amount of increasing of cheese
74  */
75 void increase_weight(cheese_data &cheese, double amount);
76
77 /**
78  * adds the cheese to the end of our dynamic array
79  *
80  * @param shop The array
81  * @param new_cheese The cheese data to add
82  */
83 void add_cheese(shop_data &shop, const cheese_data &new_cheese);
84
85 /**
86  * Return false when name is empty or price/weight is negative
87  *
88  * @param cheese cheese_data
89  * @param error_message error message to print
90  * @return true if the cheese is valid, otherwise false
91  */
92 bool cheese_valid(const cheese_data &cheese, string &error_message);
93 #endif

```

FieldGuide > 12_project > c cheese-shop.cpp

```

1 #include "cheese-shop.h"
2 #include <format>
3 #include <cmath>
4 #include <string>
5
6 using std::format;
7
8 cheese_data new_cheese(string name, double weight, int price)
9 {
10     cheese_data cheese;
11     cheese.name = name;
12     cheese.weight = weight;
13     cheese.price = price;
14     return cheese;
15 }
16
17 int total_cost(const cheese_data &cheese)
18 {
19     return round(cheese.weight * cheese.price);
20 }
21
22 string cheese_to_string(const cheese_data &cheese, bool full_details)
23 {
24     if (full_details)
25     {
26         return format("{}: {:.2f} kg, ${:.2f}, total ${:.2f}",
27             cheese.name, cheese.weight, cheese.price / 100.0, total_cost(cheese) / 100.0);
28     }
29     else
30     {
31         return cheese.name;
32     }
33 }
34
35 double reduce_weight(cheese_data &cheese, double amount)
36 {
37     if (amount <= 0)
38     {
39         return 0;
40     }
41     if (amount >= cheese.weight)
42     {
43         double removed = cheese.weight;
44         cheese.weight = 0;
45         return removed;
46     }
47     cheese.weight -= amount;
48     return amount;
49 }
50
51 void increase_weight(cheese_data &cheese, double amount)
52 {
53

```

FieldGuide > 12_project > c cheese-shop.cpp

```

54 void increase_weight(cheese_data &cheese, double amount)
55 {
56     if (amount <= 0)
57     {
58         return;
59     }
60     else
61     {
62         cheese.weight += amount;
63     }
64 }
65
66 void add_cheese(shop_data &shop, const cheese_data &new_cheese)
67 {
68     shop.cheeses.add(new_cheese);
69 }
70
71 bool cheese_valid(const cheese_data &cheese, string &error_message)
72 {
73     bool valid = true;
74     error_message = "";
75
76     if (cheese.name == "")
77     {
78         error_message = "Name is empty";
79         valid = false;
80     }
81     if (cheese.weight < 0)
82     {
83         if (error_message != "")
84             error_message += ", Weight cannot be negative";
85         else
86             error_message = "Weight cannot be negative";
87         valid = false;
88     }
89     if (cheese.price < 0)
90     {
91         if (error_message != "")
92             error_message += ", Price cannot be negative";
93         else
94             error_message = "Price cannot be negative";
95         valid = false;
96     }
97     return valid;
98 }

```

```

FIT1045
cheese-shop.h M × cheese-shop.cpp M × cheese-shop-cli.cpp M × asm3.tex × cheese-shop-test.cpp ×
FieldGuide > 12_project > C: cheese-shop-cli.cpp > handle_add_cheese(shop_data &)
1 #include "splashkit.h"
2 #include "utilities.h"
3 #include "cheese-shop.h"
4
5 #include <format>
6 using std::format;
7
8 /**
9  * compile command:
10  *
11  * clang++ -std=c++20 cheese-shop.cpp cheese-shop-cli.cpp utilities.cpp -l splashkit -o cheese-shop
12  */
13
14 /**
15  * The list of option in the main menu
16  */
17 *
18 * @option EXIT_MAIN_MENU
19 * @option ADD_CHEESE_MENU
20 * @option PRINT_STOCK_LIST_MENU
21 */
22 enum main_menu_option
23 {
24     EXIT_MAIN_MENU,
25     ADD_CHEESE_MENU,
26     PRINT_STOCK_LIST_MENU
27 };
28
29 void print_cheese(const cheese_data &cheese, bool with_full)
30 {
31     write_line(cheese_to_string(cheese, with_full));
32 }
33
34 /**
35  * Read the cheese data from user and return
36  *
37  * @return the cheese data
38  */
39 cheese_data read_cheese()
40 {
41     cheese_data cheese;
42     string error_message;
43     bool valid;
44
45     do
46     {
47         cheese.name = read_string("Enter a cheese name: ");
48         cheese.weight = read_double("Enter weight in stock (kg): ");
49         cheese.price = read_integer("Enter the price per kg (cents): ");
50
51         valid = cheese_valid(cheese, error_message);
52     }
53     while (!valid);
54
55     return cheese;
56 }
57
58 /**
59  * Perform the step need to ADD_CHEESE_MENU cheese to the shop
60  *
61  * @param shop the shop
62  */
63 void handle_add_cheese(shop_data &shop)
64 {
65     cheese_data new_cheese = read_cheese();
66     add_cheese(shop, new_cheese);
67 }
68
69 /**
70  * Show the main menu and get the option
71  *
72  * @return option
73  */
74 main_menu_option read_main_menu_option()
75 {
76     write_line("0. EXIT_MAIN_MENU");
77     write_line("1. ADD_CHEESE_MENU");
78     write_line("2. Print cheese list");
79
80     return (main_menu_option)read_integer("Select an option (0-2): ", 0, 2);
81 }
82
83 /**
84  * Output the list of stock in the shop
85  *
86  * @param shop the shop with the cheese data to output
87  */
88 void print_stock_list(const shop_data &shop)
89 {
90     if (shop.cheeses.length() == 0)
91     {
92         write_line("No cheese");
93         return;
94     }
95
96     write_line();
97     write_line("=====");
98     write_line("Cheese stock list: ");
99     write_line("=====");
100
101     for (int i = 0; i < shop.cheeses.length(); i++)
102     {
103         print_cheese(shop.cheeses[i], true);
104     }
105
106     write_line("=====");
107     write_line();
108 }
109
110 int main()
111 {
112     shop_data shop;
113     main_menu_option choice;
114
115     do
116     {
117         choice = read_main_menu_option();
118
119         switch (choice)
120         {
121             case EXIT_MAIN_MENU:
122                 write_line("EXIT_MAIN_MENUing.....");
123                 break;
124             case ADD_CHEESE_MENU:
125                 handle_add_cheese(shop);
126                 break;
127             case PRINT_STOCK_LIST_MENU:
128                 print_stock_list(shop);
129                 break;
130         }
131     }
132     while (choice != EXIT_MAIN_MENU);
133
134     return 0;
135 }
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```

- Now we want you to plan and execute a third iteration. You can choose to add anything to the program at this stage. You just need to show the following:
 - That you can design a change to the data within the program (its structs).
 - That you can add additional features to the model.
 - That you can extend the command line interface (CLI) to use your new model.

Tip: Not sure what to add? Need help thinking of a feature?

Remember you can make use of the teaching team for things like this. Learning to clarify and ask questions on expectations will help with your future studies as well.

- Describe your plan and feature extensions below.

```
> I'm planning to adding orders to shop.
```

- Describe the changes you needed to make to the data in the model:

```
> I create a struct about order_data include a dynamic array of cheese_data, then add a dynamic array about order_data into the shop_data. So, we can handle the order of multiple cheese.
```

- Describe the changes you needed to make to the functionality of the model:

```
> I have three new function:

- Add cheese to items
- Calculate the total cost of order
- Add order(items) to orders (in the shop data)

```

- Describe the changes you needed to make to the functionality of the CLI:

```
> I have three new procedure:

- read order
- Handle add order
- Print order

just like before  
And also change the enum about the menu, the read menu option, and the switch case in the main.
```

- Add a screenshot showing your progress through your iteration. Show the program running and some sample output.

>

```

h cheese-shop.h M X C++ cheese-shop.cpp M X C++ cheese-shop-cli.cpp M X C++ cheese-shop-test. ...
FieldGuide > 12_project > h cheese-shop.h > add_cheese_to_order(order_data &, const cheese_data &)
1  #ifndef CHEESE_SHOP_H
2  #define CHEESE_SHOP_H
3
4  #include "splashkit.h"
5  #include "splashkit-arrays.h"
6
7  /**
8   * Data about a cheese within the order or in a stock
9   *
10  * @field name The name of cheese
11  * @field weight The weight of cheese in kg
12  * @field price The price of cheese per kg in cents
13  */
14  struct cheese_data
15  {
16      string name;
17      double weight;
18      int price;
19  };
20
21  /**
22  * The order data about cheese_data
23  */
24  struct order_data
25  {
26      dynamic_array<cheese_data> items;
27  };
28
29  /**
30  * Data about the cheese shop
31  *
32  * @field cheeses The list of cheese in shop
33  * @field orders The list of order in the shop
34  */
35  struct shop_data
36  {
37      dynamic_array<cheese_data> cheeses;
38      dynamic_array<order_data> orders;
39  };
40
41  /**

```

```

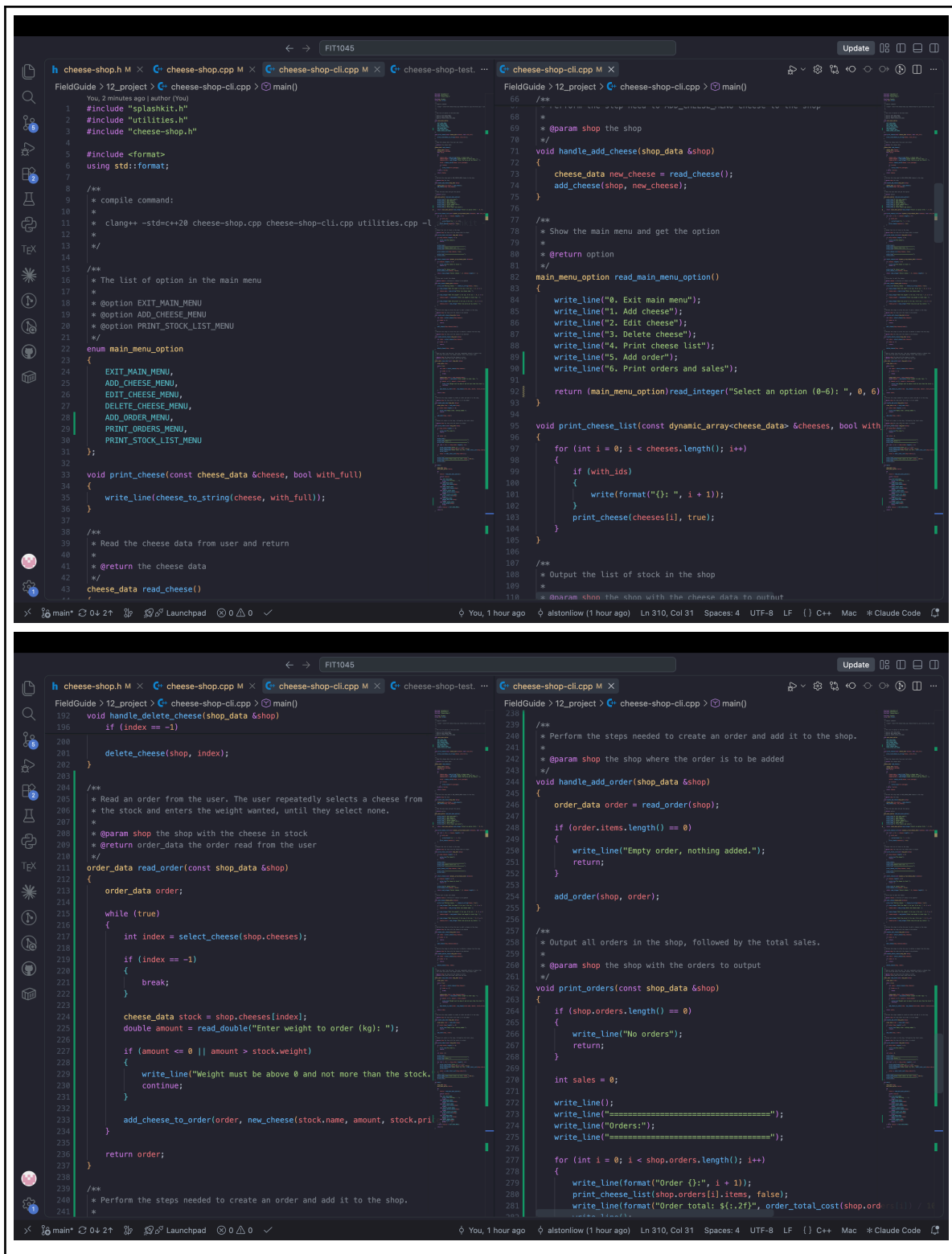
h cheese-shop.h M X
FieldGuide > 12_project > h cheese-shop.h > add_cheese_to_order(order_data &, const cheese_data &)
1  #ifndef CHEESE_SHOP_H
2  #define CHEESE_SHOP_H
3
4  /**
5   *
6   */
7  bool cheese_valid(const cheese_data &cheese, string &error_message);
8
9  /**
10  *
11  */
12  void delete_cheese(shop_data &shop, int index);
13
14  /**
15  * Add a cheese to the end of an order's item list.
16  *
17  * @param order The order to add the cheese to.
18  * @param cheese The cheese to add. Its weight is the ordered weight (kg).
19  */
20  void add_cheese_to_order(order_data &order, const cheese_data &cheese);
21
22  /**
23  * Calculate the total cost of all items in an order.
24  *
25  * @param order The order to calculate the cost of.
26  * @return int The total cost in cents.
27  */
28  int order_total_cost(const order_data &order);
29
30  /**
31  * Add an order to the end of the shop's order list.
32  *
33  * @param shop The shop to add the order to.
34  * @param order The order to add.
35  */
36  void add_order(shop_data &shop, const order_data &order);
37
38  #endif

```

```

h cheese-shop.h M X C++ cheese-shop.cpp M X C++ cheese-shop-cli.cpp M X C++ cheese-shop-test. ...
FieldGuide > 12_project > C++ cheese-shop.cpp > order_total_cost(const order_data &)
69  bool cheese_valid(const cheese_data &cheese, string &error_message)
70  {
71  }
72
73  void delete_cheese(shop_data &shop, int index)
74  {
75      if (index < 0 || index >= shop.cheeses.length())
76      {
77          return;
78      }
79      shop.cheeses.remove(index);
80  }
81
82  void add_cheese_to_order(order_data &order, const cheese_data &cheese)
83  {
84      order.items.add(cheese);
85  }
86
87  int order_total_cost(const order_data &order)
88  {
89      int total = 0;
90      for (int i = 0; i < order.items.length(); i++)
91      {
92          total += total_cost(order.items[i]);
93      }
94      return total;
95  }
96
97  void add_order(shop_data &shop, const order_data &order)
98  {
99      shop.orders.add(order);
100  }

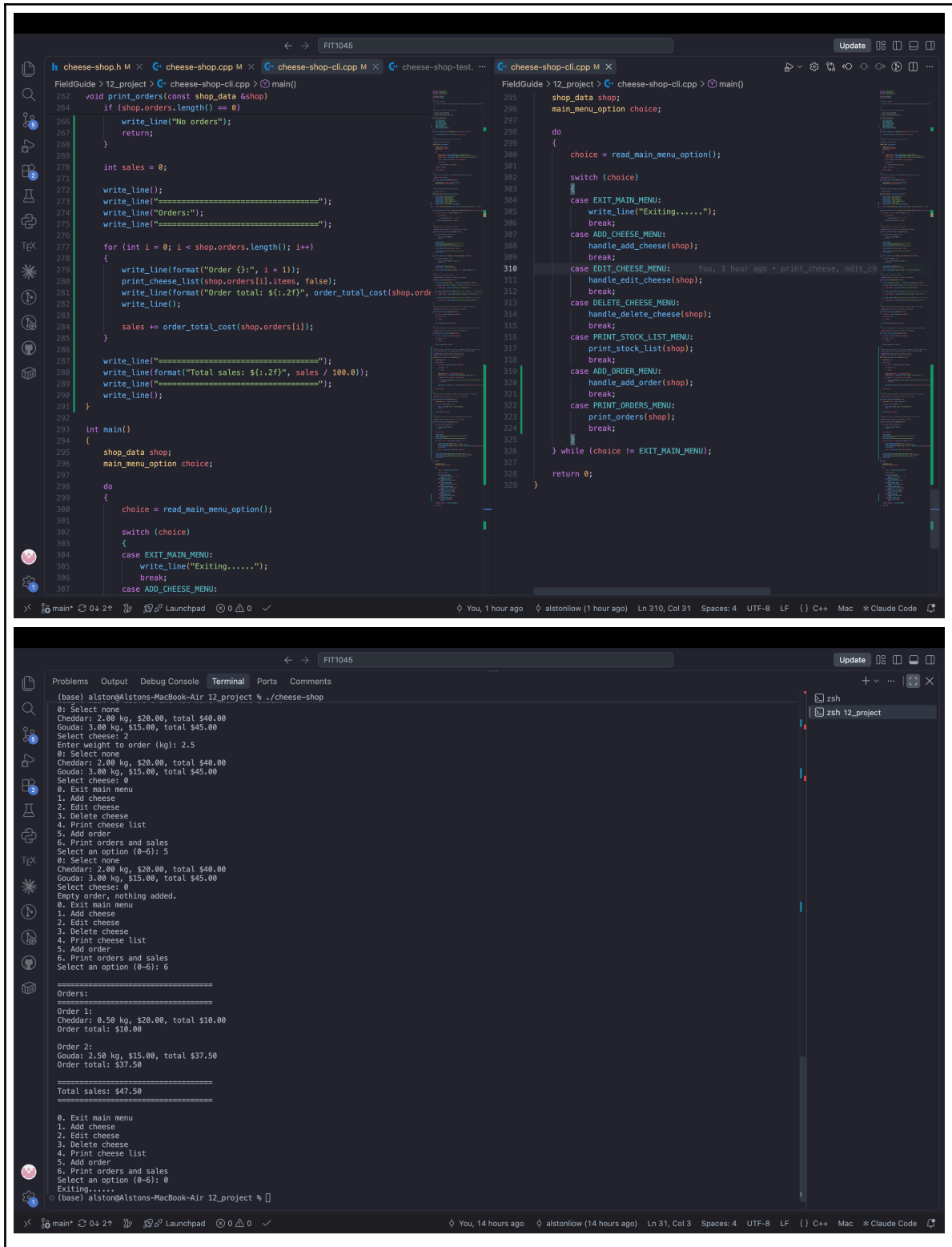
```



```

h cheese-shop.h M × C cheese-shop.cpp M × C cheese-shop-cli.cpp M × C cheese-shop-test. ... C cheese-shop-cli.cpp M ×
FieldGuide > 12_project > C cheese-shop-cli.cpp > main()
1 #include "splashkit.h"
2 #include "utilities.h"
3 #include "cheese-shop.h"
4
5 #include <format>
6 using std::format;
7
8 /**
9  * compile commands:
10  *
11  * clang++ -std=c++20 cheese-shop.cpp cheese-shop-cli.cpp utilities.cpp -l
12  *
13  */
14
15 /**
16  * The List of option in the main menu
17  *
18  * @option EXIT_MAIN_MENU
19  * @option ADD_CHEESE_MENU
20  * @option PRINT_STOCK_LIST_MENU
21  */
22 enum main_menu_option
23 {
24     EXIT_MAIN_MENU,
25     ADD_CHEESE_MENU,
26     EDIT_CHEESE_MENU,
27     DELETE_CHEESE_MENU,
28     ADD_ORDER_MENU,
29     PRINT_ORDERS_MENU,
30     PRINT_STOCK_LIST_MENU
31 };
32
33 void print_cheese(const cheese_data &cheese, bool with_full)
34 {
35     write_line(cheese_to_string(cheese, with_full));
36 }
37
38 /**
39  * Read the cheese data from user and return
40  *
41  * @return the cheese data
42  */
43 cheese_data read_cheese()
44 {
45     // ...
46 }
47
48 /**
49  * Perform the steps needed to create an order and add it to the shop.
50  *
51  * @param shop the shop where the order is to be added
52  */
53 void handle_add_order(shop_data &shop)
54 {
55     order_data order = read_order(shop);
56
57     if (order.items.length() == 0)
58     {
59         write_line("Empty order, nothing added.");
60         return;
61     }
62
63     add_order(shop, order);
64 }
65
66 /**
67  * Output all orders in the shop, followed by the total sales.
68  *
69  * @param shop the shop with the orders to output
70  */
71 void print_orders(const shop_data &shop)
72 {
73     if (shop.orders.length() == 0)
74     {
75         write_line("No orders");
76         return;
77     }
78
79     int sales = 0;
80
81     write_line();
82     write_line("=====");
83     write_line("Orders:");
84     write_line("=====");
85
86     for (int i = 0; i < shop.orders.length(); i++)
87     {
88         write_line(format("Order {}: ", i + 1));
89         print_cheese_list(shop.orders[i].items, false);
90         write_line(format("Order total: ${:.2f}", order_total_cost(shop.orders[i].items)));
91     }
92
93     write_line();
94     write_line("Total sales: ${:.2f}", sales);
95 }
96
97 /**
98  * Perform the steps needed to create an order and add it to the shop.
99  *
100  * @param shop the shop where the order is to be added
101  */
102 void handle_delete_cheese(shop_data &shop)
103 {
104     if (index == -1)
105     {
106         delete_cheese(shop, index);
107     }
108 }
109
110 /**
111  * Read an order from the user. The user repeatedly selects a cheese from
112  * the stock and enters the weight wanted, until they select none.
113  *
114  * @param shop the shop with the cheese in stock
115  * @return order_data the order read from the user
116  */
117 order_data read_order(const shop_data &shop)
118 {
119     order_data order;
120
121     while (true)
122     {
123         int index = select_cheese(shop, cheeses);
124
125         if (index == -1)
126         {
127             break;
128         }
129
130         cheese_data stock = shop.cheeses[index];
131         double amount = read_double("Enter weight to order (kg): ");
132
133         if (amount <= 0 || amount > stock.weight)
134         {
135             write_line("Weight must be above 0 and not more than the stock.");
136             continue;
137         }
138
139         add_cheese_to_order(order, new_cheese(stock.name, amount, stock.price));
140     }
141
142     return order;
143 }
144
145 /**
146  * Perform the steps needed to create an order and add it to the shop.
147  *
148  * @param shop the shop where the order is to be added
149  */
150 void handle_add_cheese(shop_data &shop)
151 {
152     // ...
153 }
154
155 /**
156  * Show the main menu and get the option
157  *
158  * @return option
159  */
160 main_menu_option read_main_menu_option()
161 {
162     write_line("0. Exit main menu");
163     write_line("1. Add cheese");
164     write_line("2. Edit cheese");
165     write_line("3. Delete cheese");
166     write_line("4. Print cheese list");
167     write_line("5. Add order");
168     write_line("6. Print orders and sales");
169
170     return (main_menu_option)read_integer("Select an option (0-6): ", 0, 6);
171 }
172
173 void print_cheese_list(const dynamic_array<cheese_data> &cheeses, bool with
174 {
175     for (int i = 0; i < cheeses.length(); i++)
176     {
177         if (with_ids)
178         {
179             write(format("{}: ", i + 1));
180         }
181         print_cheese(cheeses[i], true);
182     }
183 }
184
185 /**
186  * Output the list of stock in the shop
187  */
188 void print_stock_list(shop_data &shop)
189 {
190     // ...
191 }
192
193 /**
194  * @param shop the shop with the cheese data to output
195  */
196 void print_cheese_list(shop_data &shop)
197 {
198     // ...
199 }
200
201 /**
202  * Perform the steps needed to create an order and add it to the shop.
203  *
204  * @param shop the shop where the order is to be added
205  */
206 void handle_delete_cheese(shop_data &shop)
207 {
208     // ...
209 }
210
211 /**
212  * Read an order from the user. The user repeatedly selects a cheese from
213  * the stock and enters the weight wanted, until they select none.
214  *
215  * @param shop the shop with the cheese in stock
216  * @return order_data the order read from the user
217  */
218 order_data read_order(const shop_data &shop)
219 {
220     // ...
221 }
222
223 void print_orders(const shop_data &shop)
224 {
225     // ...
226 }
227
228 void handle_add_order(shop_data &shop)
229 {
230     // ...
231 }
232
233 void handle_add_cheese(shop_data &shop)
234 {
235     // ...
236 }
237
238 void read_main_menu_option()
239 {
240     // ...
241 }
242
243 void print_stock_list(shop_data &shop)
244 {
245     // ...
246 }
247
248 void print_cheese_list(shop_data &shop)
249 {
250     // ...
251 }
252
253 void delete_cheese(shop_data &shop, int index)
254 {
255     // ...
256 }
257
258 void add_cheese_to_order(order_data &order, cheese_data &cheese)
259 {
260     // ...
261 }
262
263 void add_order(shop_data &shop, order_data &order)
264 {
265     // ...
266 }
267
268 void print_orders(const shop_data &shop)
269 {
270     // ...
271 }
272
273 void print_cheese_list(shop_data &shop)
274 {
275     // ...
276 }
277
278 void read_main_menu_option()
279 {
280     // ...
281 }
282
283 void print_stock_list(shop_data &shop)
284 {
285     // ...
286 }
287
288 void print_cheese_list(shop_data &shop)
289 {
290     // ...
291 }
292
293 void delete_cheese(shop_data &shop, int index)
294 {
295     // ...
296 }
297
298 void add_cheese_to_order(order_data &order, cheese_data &cheese)
299 {
300     // ...
301 }
302
303 void add_order(shop_data &shop, order_data &order)
304 {
305     // ...
306 }
307
308 void print_orders(const shop_data &shop)
309 {
310     // ...
311 }
312
313 void print_cheese_list(shop_data &shop)
314 {
315     // ...
316 }
317
318 void read_main_menu_option()
319 {
320     // ...
321 }
322
323 void print_stock_list(shop_data &shop)
324 {
325     // ...
326 }
327
328 void print_cheese_list(shop_data &shop)
329 {
330     // ...
331 }
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```



The image displays two screenshots of a Visual Studio Code editor. The top screenshot shows the source code for a C++ application named 'cheese-shop'. The code is organized into two files: 'cheese-shop.h' and 'cheese-shop.cpp'. The 'cheese-shop.h' file contains a struct definition for 'shop_data' and a function 'print_orders'. The 'cheese-shop.cpp' file contains the implementation of 'print_orders' and the 'main' function. The 'main' function uses a menu-driven interface to allow users to add, edit, delete, and print orders, as well as view the total sales. The bottom screenshot shows the terminal output of the program, demonstrating its execution. The terminal shows the user interacting with the menu, adding two orders (Cheddar and Gouda), and viewing the total sales of \$47.50.

```

// cheese-shop.h
#ifndef CHEESE_SHOP_H
#define CHEESE_SHOP_H

#include <iostream>
#include <vector>
#include <string>

struct shop_data {
    std::vector<std::string> orders;
    int sales;
};

void print_orders(const shop_data &shop);

#endif

// cheese-shop.cpp
#include "cheese-shop.h"

void print_orders(const shop_data &shop) {
    if (shop.orders.length() == 0) {
        write_line("No orders");
        return;
    }

    int sales = 0;

    write_line();
    write_line("=====");
    write_line("Orders:");
    write_line("=====");

    for (int i = 0; i < shop.orders.length(); i++) {
        write_line(format("Order {}: ", i + 1));
        print_cheese_list(shop.orders[i].items, false);
        write_line(format("Order total: ${:.2f}", order_total_cost(shop.orders[i].items)));
        write_line();
        sales += order_total_cost(shop.orders[i].items);
    }

    write_line("=====");
    write_line(format("Total sales: ${:.2f}", sales / 100.0));
    write_line("=====");
}

int main() {
    shop_data shop;
    main_menu_option choice;

    do {
        choice = read_main_menu_option();

        switch (choice) {
            case EXIT_MAIN_MENU:
                write_line("Exiting.....");
                break;
            case ADD_CHEESE_MENU:
                handle_add_cheese(shop);
                break;
            case EDIT_CHEESE_MENU:
                handle_edit_cheese(shop);
                break;
            case DELETE_CHEESE_MENU:
                handle_delete_cheese(shop);
                break;
            case PRINT_STOCK_LIST_MENU:
                print_stock_list(shop);
                break;
            case ADD_ORDER_MENU:
                handle_add_order(shop);
                break;
            case PRINT_ORDERS_MENU:
                print_orders(shop);
                break;
        }
    } while (choice != EXIT_MAIN_MENU);

    return 0;
}

```

```

(base) alston@Alstons-MacBook-Air 12_project % ./cheese-shop
0: Select none
Cheddar: 2.00 kg, $20.00, total $40.00
Gouda: 3.00 kg, $15.00, total $45.00
Select cheese: 2
Enter weight to order (kg): 2.5
0: Select none
Cheddar: 2.00 kg, $20.00, total $40.00
Gouda: 3.00 kg, $15.00, total $45.00
Select cheese: 0
0: Exit main menu
1: Add cheese
2: Edit cheese
3: Delete cheese
4: Print cheese list
5: Add order
6: Print orders and sales
Select an option (0-6): 5
0: Select none
Cheddar: 2.00 kg, $20.00, total $40.00
Gouda: 3.00 kg, $15.00, total $45.00
Select cheese: 0
Empty order, nothing added.
0: Exit main menu
1: Add cheese
2: Edit cheese
3: Delete cheese
4: Print cheese list
5: Add order
6: Print orders and sales
Select an option (0-6): 6
====
Orders:
====
Order 1:
Cheddar: 0.50 kg, $20.00, total $10.00
Order total: $10.00

Order 2:
Gouda: 2.50 kg, $15.00, total $37.50
Order total: $37.50

Total sales: $47.50
====
0: Exit main menu
1: Add cheese
2: Edit cheese
3: Delete cheese
4: Print cheese list
5: Add order
6: Print orders and sales
Select an option (0-6): 0
Exiting.....
(base) alston@Alstons-MacBook-Air 12_project %

```

- Extend your glossary with a page on model separated design.

Reflect:

- What was the most interesting thing you learnt through this task, and why?

> The most interesting thing is using the separate file to write a program. Include a model (.h and .cpp) and a command line interface (CLI). It is useful because when the program become more complex and significant, we also can separate the model file into different function file, using this concept, the program is easy to iterate and maintain.

- How much of your model would be useful in a graphical user interface version of your program? Could the model also work as the code within a server for a web application? What does this tell you about the point of this task?

> I think my model can reuse in the GUI version because it just contain the logic, but need to rewrite a program like the CLI one, to interface in GUI version.

For the server or web, I think the model also can reuse. But the issue is my model cannot store the data, so maybe need to re-enter all the data every time.

I think the point of this task is separate of model or MVC (Model-View-Controller).

- Looking back, how long did it take to complete this task? What did you spend the most time on?

> 12 hours. Take most time on iterating the program because need change many thing.

Disclose:

- Explain your use of Generative AI for this task. If you didn't use it, just write "I did not use any AI for this task". If you did use it, please state which AI model you used (e.g. ChatGPT) and how.

> I did not use any AI for this task.

Submit:

- Export this worksheet as a PDF.
- Upload your submission to OnTrack: